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Flux Advanced Mathematics

26 August 2026

Morning / Afternoon

Paper reference **FLX/06**

Advanced Extension Award Mathematics

PRACTICE PAPER

You must have:
Mathematical Formulae and Statistical Tables

Total Marks
61

Instructions

Use black ink or ball-point pen.
Fill in the boxes at the top of this page with your name and candidate number.
Answer **all** questions.
Answer the questions in the spaces provided.
Calculators may not be used.
You must **show all your working.**

Information

The total mark for this paper is 61.
There are 6 questions in this question paper.
Questions collected from Edexcel 2010 P1, Edexcel 2019 P1, Edexcel 2017 P1, Edexcel 2014 P1.

Advice

Read each question carefully before you start to answer it.
Try to answer every question.
Check your answers if you have time at the end.

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1. Edexcel 2010 P1 Q4 **(16)**

VECTORS · SCALAR PRODUCT

Figure 1 *(A rectangular cuboid OABCDEFG is shown. O is the origin, A is on the x-axis, C is on the y-axis, and D is on the z-axis. The vertices are given as vectors in the text).* Figure 1 shows a cuboid $OABCDEFG$, where O is the origin, A has position vector $5\mathbf{i}$, C has position vector $10\mathbf{j}$ and D has position vector $20\mathbf{k}$.

(a) Find the cosine of angle CAF. **(4)**

Given that the point X lies on AC and that FX is perpendicular to AC, **(b)** find the position vector of point X and the distance FX. **(7)**

The line l_1 passes through O and through the midpoint of the face $ABFE$. The line l_2 passes through A and through the midpoint of the edge FG. **(c)** Show that l_1 and l_2 intersect and find the coordinates of the point of intersection. **(5)** **(Total 16 marks)**

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2. Edexcel 2010 P1 Q6 **(10)**

DIFFERENTIATION · STATIONARY POINTS

(a) Given that $x^4 + y^4 = 1$, prove that $x^2 + y^2$ is a maximum when $x = \pm y$, and find the maximum and minimum values of $x^2 + y^2$. **(7)**

(b) On the same diagram, sketch the curves C_1 and C_2 with equations $x^4 + y^4 = 1$ and $x^2 + y^2 = 1$ respectively. **(2)**

(c) Write down the equation of the circle C_3, centre the origin, which touches the curve C_1 at the points where $x = \pm y$. **(1)** **(Total 10 marks)**

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3. Edexcel 2019 P1 Q6 **(—)**

DIFFERENTIATION · THE PRODUCT RULE

Figure 1 *(A sketch of part of the curve $y = x\sin(\ln x)$ for $x \geq 1$. The curve starts at the origin, arches upwards, and crosses the x-axis at a point A. The area under the curve between $x = 1$ and $x = A$ is shaded and labeled R).*

Figure 1 shows a sketch of part of the curve with equation

$$y = x\sin(\ln x) \quad x \geq 1$$

For $x > 1$, the curve first crosses the x-axis at the point A.

(a) Find the x coordinate of A. **(3)**

(b) Differentiate $x\sin(\ln x)$ and $x\cos(\ln x)$ with respect to x and hence find

$$\int \sin(\ln x)dx$$

and

$$\int \cos(\ln x)dx$$

(7)

(c) (i) Find

$$\int x\sin(\ln x)dx$$

. (ii) Hence show that the area of the shaded region R, bounded by the curve and the x-axis between the points $(1, 0)$ and A, is $\frac{1}{5}(e^{2\pi} + 1)$ **(9)**

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4. Edexcel 2010 P1 Q5 **(12)**

INTEGRATION · INTEGRATION BY SUBSTITUTION

$$I = \int \frac{1}{(x-1)\sqrt{x^2-1}} dx, \quad x > 1$$

(a) Use the substitution $x = 1 + u^{-1}$ to show that $I = -\left(\frac{x+1}{x-1}\right)^{\frac{1}{2}} + c$ **(7)**

(b) Hence show that

$$\int_{\sec\alpha}^{\sec\beta} \frac{1}{(x-1)\sqrt{x^2-1}} dx = \cot\left(\frac{\alpha}{2}\right) - \cot\left(\frac{\beta}{2}\right), \quad 0 < \alpha < \beta < \frac{\pi}{2}$$

(5) (Total 12 marks)

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5. Edexcel 2017 P1 Q7 **(—)**

INTEGRATION · AREAS BETWEEN CURVES AND LINES

Figure 3 *(A sketch showing part of the curve C and a tangent line L. The line touches the curve in two places, forming two distinct local minima contacts labeled P and Q. A shaded region R is trapped between the curve and the line between these two touching points).*

Figure 3 shows part of the curve C with equation

$$y = x^4 - 10x^3 + 33x^2 - 34x$$

and the line L with equation $y = mx + c$. The line L touches C at the points P and Q with x coordinates p and q respectively.

(a) Explain why

$$x^4 - 10x^3 + 33x^2 - (34 + m)x - c = (x - p)^2(x - q)^2$$

(2)

The finite region R, shown shaded in Figure 3, is bounded by C and L. **(b)** Use integration by parts to show that the area of R is

$$\frac{(q - p)^5}{30}$$

(6)

(c) Show that

$$(x - p)^2(x - q)^2 = x^4 - 2(p + q)x^3 + Sx^2 - Tx + U$$

where S, T and U are expressions to be found in terms of p and q. **(5)**

(d) Using part (a) and part (c) find the value of p, the value of q and the equation of L. **(8)**

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6. Edexcel 2014 P1 Q7 **(23)**

INTEGRATION · AREAS UNDER CURVES

Figure 2 *(A circular tower C is centered at T(0,1) with radius 1. A goat attached to the origin O(0,0) via a string of length π roams the region outside. The string wraps around the tower up to point A, then runs straight to the goat at G(x,y)).*

A circular tower stands in a large horizontal field of grass. A goat is attached to one end of a string and the other end of the string is attached to the fixed point O at the base of the tower. Taking the point O as the origin $(0, 0)$, the centre of the base of the tower is at the point $T(0, 1)$. The radius of the base of the tower is 1. The string has length π and you may ignore the size of the goat. The curve C represents the edge of the region that the goat can reach as shown in Figure 2.

(a) Write down the equation of C for $y < 0$. **(1)**

When the goat is at the point $G(x, y)$ with $x > 0$ and $y > 0$, as shown in Figure 2, the string lies along OAG where OA is an arc of the circle with angle $OTA = \theta$ radians and AG is a tangent to the circle at A.

(b) With the aid of a suitable diagram show that $x = \sin\theta + (\pi - \theta)\cos\theta$ $y = 1 - \cos\theta + (\pi - \theta)\sin\theta$ **(5)**

(c) By considering

$$\int y \frac{dx}{d\theta} d\theta$$

, show that the area between C, the positive x-axis and the positive y-axis can be expressed in the form

$$\int_0^\pi u \sin u du + \int_0^\pi u^2 \sin^2 u du + \int_0^\pi u \sin u \cos u du$$

(5)

(d) Show that

$$\int_0^\pi u^2 \sin^2 u du = \frac{\pi^3}{6} + \int_0^\pi u \sin u \cos u du$$

(4)

(e) Hence find the area of grass that can be reached by the goat. **(8)** **(Total 23 marks)**

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Mark Schemes

Question 1 — Mark Scheme

Part (a)

Step 1: Find vectors AC and AF $A = (5, 0, 0)$, $C = (0, 10, 0)$, $F = (5, 10, 20)$.

$$\vec{AC} = \vec{OC} - \vec{OA} = \begin{pmatrix} -5 \\ 10 \\ 0 \end{pmatrix}$$

$$\vec{AF} = \vec{OF} - \vec{OA} = \begin{pmatrix} 0 \\ 10 \\ 20 \end{pmatrix}$$

[B1] Correct vectors.

Step 2: Calculate magnitudes $|\vec{AC}| = \sqrt{25 + 100} = \sqrt{125}$ $|\vec{AF}| = \sqrt{100 + 400} = \sqrt{500}$

[B1] Correct moduli.

Step 3: Use dot product for angle $\vec{AC} \cdot \vec{AF} = (-5)(0) + (10)(10) + (0)(20) = 100$

$$\cos(\angle CAF) = \frac{100}{\sqrt{125}\sqrt{500}} = \frac{100}{250} = \frac{2}{5} = 0.4$$

[M1, A1] M1 for complete method for $\cos$. A1 for 0.4 or $2/5$.

Part (b)

Step 4: Form equation for line AC

$$\vec{OX} = \vec{OA} + t\vec{AC} = \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -5 \\ 10 \\ 0 \end{pmatrix} = \begin{pmatrix} 5 - 5t \\ 10t \\ 0 \end{pmatrix}$$

[M1] Attempt equation for AC or variable OX.

Step 5: Form vector FX

$$\vec{FX} = \vec{OX} - \vec{OF} = \begin{pmatrix} 5 - 5t - 5 \\ 10t - 10 \\ 0 - 20 \end{pmatrix} = \begin{pmatrix} -5t \\ 10t - 10 \\ -20 \end{pmatrix}$$

[M1] Attempt FX in terms of one unknown.

Step 6: Apply perpendicularity condition

$$\vec{FX} \cdot \vec{AC} = 0 \Rightarrow \begin{pmatrix} -5t \\ 10t - 10 \\ -20 \end{pmatrix} \cdot \begin{pmatrix} -5 \\ 10 \\ 0 \end{pmatrix} = 0$$

$$25t + 100t - 100 = 0 \Rightarrow 125t = 100 \Rightarrow t = 0.8$$

[M1, A1] Correct use of dot product to get linear equation in t; $t = 0.8$.

Step 7: Find X and FX distance

$$\vec{OX} = \begin{pmatrix} 5-4 \\ 8 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 8 \\ 0 \end{pmatrix}$$

$$\vec{FX} = \begin{pmatrix} -4 \\ -2 \\ -20 \end{pmatrix}$$

$|\vec{FX}| = \sqrt{16 + 4 + 400} = \sqrt{420}$

[A1, M1, A1] A1 for $\vec{OX}$. M1 for attempt at $|FX|$. A1 for $\sqrt{420}$.

Part (c)

Step 8: Define lines l1 and l2 Midpoint of $ABFE$:

$$\frac{1}{2}(\vec{OA} + \vec{OF}) = \frac{1}{2}((5,0,0) + (5,10,20)) = (5,5,10)$$

.

$$l_1 : \mathbf{r} = \lambda \begin{pmatrix} 5 \\ 5 \\ 10 \end{pmatrix}$$

Midpoint of FG:

$$\frac{1}{2}(\vec{OF} + \vec{OG}) = \frac{1}{2}((5,10,20) + (0,10,20)) = (2.5,10,20)$$

.

$$l_2 : \mathbf{r} = \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix} + \mu \left(\begin{pmatrix} 2.5 \\ 10 \\ 20 \end{pmatrix} - \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix} \right) = \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -2.5 \\ 10 \\ 20 \end{pmatrix}$$

[B1, B1] Correct vector equations for both lines.

Step 9: Solve for intersection Equate components: $5\lambda = 5 - 2.5\mu$ $5\lambda = 10\mu$ $10\lambda = 20\mu$

[M1] Clear attempt to solve.

$$10\mu = 5 - 2.5\mu \Rightarrow 12.5\mu = 5 \Rightarrow \mu = 0.4$$

$5\lambda = 4 \Rightarrow \lambda = 0.8$

[A1] Correct parameters.

Step 10: Find intersection point Check in z: $10(0.8) = 8$ and $20(0.4) = 8$ (consistent). Point is $0.8 \times (5,5,10) = (4,4,8)$.

[A1] Correct intersection coordinates. (S+ for clear attempt to check intersection consistency).

Question 2 — Mark Scheme

Part (a)

Step 1: Set up the area/value function Let $A = x^2 + y^2$. From $x^4 + y^4 = 1$, we have $y^2 = \sqrt{1 - x^4}$. $A = x^2 + \sqrt{1 - x^4}$

[B1] Expression as a function of x only.

Step 2: Differentiate

$$\frac{dA}{dx} = 2x + \frac{1}{2}(1 - x^4)^{-1/2}(-4x^3) = 2x - \frac{2x^3}{\sqrt{1 - x^4}}$$

[M1, A1] Some correct differentiation.

Step 3: Find turning points Set derivative to zero:

$$2x = \frac{2x^3}{\sqrt{1 - x^4}} \Rightarrow \sqrt{1 - x^4} = x^2 \quad (\text{for } x \neq 0)$$

$1 - x^4 = x^4 \Rightarrow x^4 = \frac{1}{2}$

[M1, M1] Reaching $x^4 = 1/2$ (or $x^2 = y^2 \Rightarrow x = \pm y$).

Step 4: Find max/min values If $x^4 = 1/2$, then $y^4 = 1/2$. $A = x^2 + y^2 = \sqrt{1/2} + \sqrt{1/2} = \sqrt{2}$. This is the maximum. If $x = 0$, then $y^4 = 1 \Rightarrow y = \pm 1$. $A = 1$. This is the minimum.

[B1, B1] Max is $\sqrt{2}$, Min is 1.

Part (b)

Step 5: Sketch the curves $x^2 + y^2 = 1$ is a standard unit circle. $x^4 + y^4 = 1$ resembles a square with rounded corners ("squircle"). It passes through $(\pm 1, 0)$ and $(0, \pm 1)$ exactly on the unit circle, but bulges outwards elsewhere, reaching a maximum distance of $\sqrt[4]{1/2}$ from the origin along the lines $y = \pm x$. Wait, maximum distance squared is $\sqrt{2}$, so distance is $2^{1/4} \approx 1.189$. So it's outside the circle.

[B1] Circle, centre $(0, 0)$, $r = 1$.

[B1] Squircle curve outside the unit circle, touching at the axes.

Part (c)

Step 6: Circle C3 The circle C_3 touches C_1 at its furthest points $x = \pm y$. The radius squared here is $\sqrt{2}$ (found in part a). Equation: $x^2 + y^2 = \sqrt{2}$.

[B1] Correct equation.

Question 3 — Mark Scheme

Part (a)

Step 1: Find root A Set $y = 0$: $x \sin(\ln x) = 0$. Since $x \geq 1$, we need $\sin(\ln x) = 0$.

$$\Rightarrow \ln x = \pi$$

(the first positive root greater than 0, corresponding to $x > 1$).

[M1, M1] Sets equal to 0 and extracts the second solution for sine (first is 0 at $x = 1$).

Step 2: Calculate x $x = e^{\pi}$

[A1] Exact value for x.

Part (b)

Step 3: Differentiate first expression

$$\frac{d}{dx}(x\sin(\ln x)) = 1\cdot\sin(\ln x) + x\cdot\cos(\ln x)\cdot\frac{1}{x} = \sin(\ln x) + \cos(\ln x)$$

[M1, A1] Use of the Product Rule. All correct, simplified.

Step 4: Differentiate second expression

$$\frac{d}{dx}(x\cos(\ln x)) = 1\cdot\cos(\ln x) + x\cdot(-\sin(\ln x))\cdot\frac{1}{x} = \cos(\ln x) - \sin(\ln x)$$

[M1, A1] Product Rule used. All correct, simplified.

Step 5: Integrate to find primitives Integrating the results:

$$x\sin(\ln x) = \int \sin(\ln x)dx + \int \cos(\ln x)dx$$

$$x\cos(\ln x) = \int \cos(\ln x)dx - \int \sin(\ln x)dx$$

Adding the two equations:

$$x\sin(\ln x) + x\cos(\ln x) = 2\int \cos(\ln x)dx \Rightarrow \int \cos(\ln x)dx = \frac{1}{2}x(\sin(\ln x) + \cos(\ln x))$$

Subtracting the equations:

$$x\sin(\ln x) - x\cos(\ln x) = 2\int \sin(\ln x)dx \Rightarrow \int \sin(\ln x)dx = \frac{1}{2}x(\sin(\ln x) - \cos(\ln x))$$

[M1, A1, A1] Must be a genuine attempt combining the derivatives to find the integrals. Correct answers.

Part (c)

Step 6: Integrate x sin(ln x) by parts (i) Let

$$S = \int x\sin(\ln x)dx$$

. Using integration by parts:

$$u = \sin(\ln x) \Rightarrow du = \cos(\ln x)\cdot\frac{1}{x}dx$$

$$dv = xdx \Rightarrow v = \frac{1}{2}x^2$$

$$S = \frac{1}{2}x^2\sin(\ln x) - \int \frac{1}{2}x\cos(\ln x)dx$$

[M1, A1] Application of parts.

Step 7: Second application of parts Let

$$C = \int x \cos(\ln x) dx$$

.

$$C = \frac{1}{2}x^2 \cos(\ln x) - \int \frac{1}{2}x(-\sin(\ln x))dx = \frac{1}{2}x^2 \cos(\ln x) + \frac{1}{2}S$$

[M1, A1] Applies parts a second time on C.

Step 8: Solve for S Substitute C back into S:

$$S = \frac{1}{2}x^2 \sin(\ln x) - \frac{1}{2}\left(\frac{1}{2}x^2 \cos(\ln x) + \frac{1}{2}S\right)$$

$$S = \frac{1}{2}x^2 \sin(\ln x) - \frac{1}{4}x^2 \cos(\ln x) - \frac{1}{4}S$$

[M1] Solves simultaneously to eliminate the integral "loop".

$$\frac{5}{4}S = \frac{1}{4}x^2(2\sin(\ln x) - \cos(\ln x))$$

$$\Rightarrow S = \frac{1}{5}x^2(2\sin(\ln x) - \cos(\ln x))(+k)$$

[M1, A1] Rearranges to find S and obtains correct expression.

Step 9: Evaluate definite integral (ii) Apply limits from $x = 1$ to $x = e^\pi$:

$$\text{Area} = \left[\frac{1}{5}x^2(2\sin(\ln x) - \cos(\ln x))\right]_1^{e^\pi}$$

Upper limit e^π:

$$\frac{1}{5}(e^\pi)^2(2\sin(\pi) - \cos(\pi)) = \frac{1}{5}e^{2\pi}(0 - (-1)) = \frac{1}{5}e^{2\pi}$$

Lower limit 1:

$$\frac{1}{5}(1)^2(2\sin(0) - \cos(0)) = \frac{1}{5}(0 - 1) = -\frac{1}{5}$$

[M1] Uses limits $(1, e^\pi)$ in their S.

Step 10: Final Area

$$\text{Area} = \frac{1}{5}e^{2\pi} - \left(-\frac{1}{5}\right) = \frac{1}{5}(e^{2\pi} + 1)$$

[A1] Correct Given Answer from fully correct working.

Question 4 — Mark Scheme

Part (a)

Step 1: Apply substitution $x = 1 + u^{-1} \Rightarrow \frac{dx}{du} = -u^{-2}$

[B1] Correct dx/du.

Step 2: Transform integral into u $x - 1 = u^{-1}$

$$x^2 - 1 = (1 + u^{-1})^2 - 1 = 1 + 2u^{-1} + u^{-2} - 1 = 2u^{-1} + u^{-2}$$

$$I = \int \frac{1}{u^{-1}\sqrt{2u^{-1} + u^{-2}}}(-u^{-2})du$$

[M1] Attempt to get I in u only.

Step 3: Simplify integrand

$$I = \int \frac{-u^{-2}}{u^{-1}\sqrt{u^{-2}(2u+1)}}du = \int \frac{-u^{-2}}{u^{-2}\sqrt{2u+1}}du = -\int \frac{1}{\sqrt{2u+1}}du$$

[A1] Correct simplified expression.

Step 4: Integrate

$$I = -\int (1 + 2u)^{-1/2}du = -(1 + 2u)^{1/2} + c$$

[M1, A1] Attempt to integrate; correct integration.

Step 5: Convert back to x Substitute $u = \frac{1}{x-1}$:

$$I = -\left(1 + \frac{2}{x-1}\right)^{1/2} + c = -\left(\frac{x - 1 + 2}{x - 1}\right)^{1/2} + c = -\left(\frac{x+1}{x-1}\right)^{\frac{1}{2}} + c$$

[M1, A1 cso] Substitute u back; correct final expression including $+c$.

Part (b)

Step 6: Evaluate integral with limits

$$[I]_{\sec\alpha}^{\sec\beta} = -\left(\frac{\sec\beta + 1}{\sec\beta - 1}\right)^{\frac{1}{2}} - \left(-\left(\frac{\sec\alpha + 1}{\sec\alpha - 1}\right)^{\frac{1}{2}}\right)$$

[M1] Use of part (a).

Step 7: Simplify trig fractions Multiply numerator and denominator by $\cos\beta$:

$$\frac{\sec\beta + 1}{\sec\beta - 1} = \frac{1 + \cos\beta}{1 - \cos\beta}$$

[M1] Convert to $\cos$.

Step 8: Use half-angle identities

$$\frac{1 + \cos\beta}{1 - \cos\beta} = \frac{2\cos^2(\beta/2)}{2\sin^2(\beta/2)} = \cot^2\left(\frac{\beta}{2}\right)$$

[M1] Use of half angle formulae.

Step 9: Conclude

$$\sqrt{\cot^2\left(\frac{\beta}{2}\right)} = \cot\left(\frac{\beta}{2}\right)$$

(since $0 < \beta/2 < \pi/4$, cot is positive).

$$= -\cot\left(\frac{\beta}{2}\right) + \cot\left(\frac{\alpha}{2}\right) = \cot\left(\frac{\alpha}{2}\right) - \cot\left(\frac{\beta}{2}\right)$$

[M1, A1 cso] Correct removal of square root and final result.

Question 5 — Mark Scheme

Part (a)

Step 1: Set up intersection The intersection of the curve C and line L is found by equating them:

$$x^4 - 10x^3 + 33x^2 - 34x = mx + c \Rightarrow x^4 - 10x^3 + 33x^2 - (34+m)x - c = 0$$

[B1] Reason for LHS (Curve - Line).

Step 2: Argue tangency and roots Because the line L is a tangent to C at points $x = p$ and $x = q$, these roots must be repeated (double roots). As the leading coefficient is 1 and it is a quartic, the polynomial factors exactly as

$$(x-p)^2(x-q)^2$$

.

[B1] Mention of double roots and reason for squares balancing powers.

Part (b)

Step 3: Set up integral for Area R

$$\text{Area} = \int_p^q (L - C)dx = \int_p^q -(x-p)^2(x-q)^2 dx \quad \text{or} \quad \int_p^q (x-p)^2(x-q)^2 dx$$

We evaluate

$$\int_p^q (x-p)^2(x-q)^2 dx$$

(area is the positive value of this). Let $u = (x-p)^2$ and $dv = (x-q)^2 dx$.

[M1] Attempt 1st step of integration by parts.

Step 4: First Integration by Parts

$$v = \frac{(x-q)^3}{3}, \quad du = 2(x-p)dx$$

$$\int_p^q (x-p)^2(x-q)^2 dx = \left[(x-p)^2 \frac{(x-q)^3}{3}\right]_p^q - \int_p^q 2(x-p)\frac{(x-q)^3}{3} dx$$

The boundary term evaluates to 0 at both q and p.

[A1] Correctly gets 1st boundary integral = 0.

[A1] Correct 2nd integral expression.

Step 5: Second Integration by Parts For

$$-\int_p^q \frac{2}{3}(x-p)(x-q)^3 dx$$

: Let $u = (x - p)$ and $dv = (x - q)^3 dx$.

$$v = \frac{(x-q)^4}{4}, \quad du = dx$$

$$= -\frac{2}{3}\left(\left[(x-p)\frac{(x-q)^4}{4}\right]_p^q - \int_p^q \frac{(x-q)^4}{4}dx\right)$$

[M1] Attempt 2nd step of integration by parts.

Step 6: Final Evaluation The boundary term again evaluates to 0.

$$= \frac{2}{3}\int_p^q \frac{(x-q)^4}{4}dx = \frac{1}{6}\int_p^q (x-q)^4 dx$$

= \frac{1}{6} \left[\frac{(x - q)^5}{5} \right]_p^q = \frac{1}{30} \left(0 - (p - q)^5 \right)$ $= \frac{-(p - q)^5}{30} = \frac{(q - p)^5}{30} \quad (*)

[A1, A1 cso] Correct work leading to the single integral with zeros seen. No incorrect working seen leading to this.

Part (c)

Step 7: Expand the squared terms

$$(x-p)^2(x-q)^2 = (x^2 - 2px + p^2)(x^2 - 2qx + q^2)$$

[M1] Starts the expansion multiplying the two quadratics.

Step 8: Collect terms

$$= x^4 - 2qx^3 + q^2x^2 - 2px^3 + 4pqx^2 - 2pq^2x + p^2x^2 - 2p^2qx + p^2q^2$$

$$= x^4 - 2(p+q)x^3 + (p^2 + q^2 + 4pq)x^2 - (2pq^2 + 2p^2q)x + p^2q^2$$

[A1] First two terms correct (cso).

[A1] Correct expression for $S = p^2 + q^2 + 4pq$.

[A1] Correct expression for $T = 2pq(p+q)$ (or equivalent).

[A1] Correct expression for $U = p^2q^2$.

Part (d)

Step 9: Compare coefficients for x^3 and x^2 From parts (a) and (c), comparing coefficients:

$$-2(p+q) = -10 \Rightarrow p + q = 5$$

$$S = 33 \Rightarrow p^2 + q^2 + 4pq = 33$$

[B1] Extracts $p + q = 5$.

Step 10: Solve for p and q We know

$$(p+q)^2 = p^2 + q^2 + 2pq = 25.$$

Thus

$$(p^2 + q^2 + 4pq) - (p^2 + q^2 + 2pq) = 33 - 25 \Rightarrow 2pq = 8 \Rightarrow pq = 4.$$

Substitute $p = 5 - q$:

$$q(5 - q) = 4 \Rightarrow q^2 - 5q + 4 = 0.$$

[M1] Uses second equation and $p + q = 5$ substitution.
[M1] Solves the resulting quadratic in one variable.

$$(q - 4)(q - 1) = 0 \Rightarrow q = 1 \text{ or } q = 4$$

[A1] Finds roots 1 and 4. From the diagram, Q is to the right of P, so $q > p$. Thus $p = 1$ and $q = 4$.
[A1] Correctly identifies p and q.
Step 11: Determine m and c We have

$$T = 2pq(p + q) = 2(4)(5) = 40$$

. From (a), the x coefficient is $-(34 + m)$. From (c), it is $-T$.

$$34 + m = 40 \Rightarrow m = 6$$

[M1, A1] Uses T to find m.
We have

$$U = p^2q^2 = 1^2 \times 4^2 = 16.$$

From (a), the constant term is $-c$. From (c), it is U.

$$-c = 16 \Rightarrow c = -16$$

Equation of line L: $y = 6x - 16$
[A1] Correct line equation.

Question 6 — Mark Scheme

Part (a)

Step 1: Write equation for y < 0 When $y < 0$, the string does not wrap around the tower, so the region is a simple semicircle of radius π centered at the origin. $x^2 + y^2 = \pi^2$
[B1] Correct equation.

Part (b)

Step 2: Define components of the path The arc length $OA = r\theta = (1)\theta = \theta$. The remaining string length is $AG = \pi - \theta$.
[B1, B1] $OA = \theta$ and $AG = \pi - \theta$.
Step 3: Express coordinates of A T is at $(0, 1)$. Relative to T, the position of A is given by rotating the vector $(0, -1)$ by θ counterclockwise (based on the diagram, the angle θ is measured from the negative y-axis). Thus,

$$A = (1 \sin\theta, 1 - 1\cos\theta) = (\sin\theta, 1 - \cos\theta)$$

.

[M1] Clear method for calculating coordinate components.

Step 4: Vector AG Since AG is a tangent at A, it is perpendicular to TA. The direction of TA is $(\sin\theta, -\cos\theta)$. The tangent direction (moving away from the circle) is $(\cos\theta, \sin\theta)$. The vector $\overrightarrow{AG}$ has length $\pi - \theta$ and direction $(\cos\theta, \sin\theta)$. Thus,

$$\overrightarrow{AG} = ((\pi - \theta)\cos\theta, (\pi - \theta)\sin\theta)$$

.

[A1, A1 cso] Fully correct components added to coordinates of A to yield the final given answers: $x = \sin\theta + (\pi - \theta)\cos\theta$ $y = 1 - \cos\theta + (\pi - \theta)\sin\theta$

Part (c)

Step 5: Differentiate x with respect to theta

$$\frac{dx}{d\theta} = \cos\theta - 1 \cdot \cos\theta - (\pi - \theta)\sin\theta = -(\pi - \theta)\sin\theta$$

[M1, A1] Application of product rule; correct derivative.

Step 6: Set up Area Integral The area in the first quadrant is given by

$$\int_{x=0}^{x=X_{max}} y dx = \int_{\theta_{max}}^{\theta_{min}} y \frac{dx}{d\theta} d\theta$$

. At $y = 0$, the string is fully unwrapped along the x-axis, so $\theta = \pi$. At $x = 0$, the string points straight up, $\theta = 0$.

$$\text{Area} = \int_{\pi}^{0} [1 - \cos\theta + (\pi - \theta)\sin\theta][-(\pi - \theta)\sin\theta] d\theta$$

[M1] Ignores limits for the mark, correct integrand structure.

Step 7: Apply Substitution Let $u = \pi - \theta \Rightarrow d\theta = -du$. Limits: $\theta = \pi \Rightarrow u = 0$; $\theta = 0 \Rightarrow u = \pi$. Also, $\cos(\pi - u) = -\cos u$ and $\sin(\pi - u) = \sin u$.

$$\text{Area} = \int_{0}^{\pi} [1 - (-\cos u) + u\sin u][-u\sin u](-du)$$

[M1] Suitable substitution into integral.

Step 8: Expand into requested format

$$= \int_{0}^{\pi} [1 + \cos u + u\sin u][u\sin u] du$$

$$= \int_{0}^{\pi} (u\sin u + u\cos u\sin u + u^2\sin^2 u) du \quad (*)$$

[A1 cso] Correctly simplified to printed answer.

Part (d)

Step 9: Use integration by parts on the second term

$$\int_{0}^{\pi} u^2\sin^2 u du = \int_{0}^{\pi} u^2 \left(\frac{1 - \cos 2u}{2}\right) du = \int_{0}^{\pi} \frac{u^2}{2} du - \int_{0}^{\pi} \frac{u^2}{2}\cos 2u du$$

[M1] Uses double angle identity.

$$\int_0^\pi \frac{u^2}{2} du = \left[\frac{u^3}{6}\right]_0^\pi = \frac{\pi^3}{6}$$

[A1] Correct evaluation of the first part.

Step 10: Apply parts to the second half For

$$\int \frac{u^2}{2} \cos 2u du$$

: Let $v = u^2/2 \Rightarrow dv = udu$. Let $dw = \cos 2udu \Rightarrow w = \frac{\sin 2u}{2}$.

$$= \left[\frac{u^2}{4} \sin 2u\right]_0^\pi - \int_0^\pi u \frac{\sin 2u}{2} du$$

[M1] Correct application of integration by parts.

Since $\sin(2\pi) = \sin(0) = 0$, the boundary term vanishes. Substitute $\sin 2u = 2 \sin u \cos u$:

$$= 0 - \int_0^\pi u \frac{2 \sin u \cos u}{2} du = - \int_0^\pi u \sin u \cos udu$$

Thus:

$$\int_0^\pi u^2 \sin^2 udu = \frac{\pi^3}{6} - \left(- \int_0^\pi u \sin u \cos udu\right) = \frac{\pi^3}{6} + \int_0^\pi u \sin u \cos udu \quad (*)$$

[A1 cso] Completes proof showing boundary evaluates to 0.

Part (e)

Step 11: Integrate remaining components We need to calculate the area from (c).

Substitute the result from (d):

$$\text{Area (1st quadrant)} = \int_0^\pi u \sin udu + \left(\frac{\pi^3}{6} + \int_0^\pi u \sin u \cos udu\right) + \int_0^\pi u \sin u \cos udu$$

$$= \frac{\pi^3}{6} + \int_0^\pi u \sin udu + 2 \int_0^\pi u \sin u \cos udu$$

Step 12: Evaluate integrals For

$$\int u \sin udu$$

: Parts: $v = u$, $dw = \sin udu \Rightarrow w = - \cos u$.

$$= [-u \cos u]_0^\pi + \int_0^\pi \cos udu = -\pi(-1) - 0 + [\sin u]_0^\pi = \pi$$

[M1, A1] Attempts by parts; correct evaluation $= \pi$.

For

$$\int u \sin u \cos u du = \int u \frac{\sin 2u}{2} du$$

: Parts: $v = u/2$, $dw = \sin 2u du \Rightarrow w = -\frac{\cos 2u}{2}$.

$$= \left[-u\frac{\cos 2u}{4}\right]_0^\pi + \int_0^\pi \frac{\cos 2u}{4} du = -\frac{\pi(1)}{4} - 0 + \left[\frac{\sin 2u}{8}\right]_0^\pi = -\frac{\pi}{4}$$

[M1, A1] Attempts by parts; correct evaluation $= -\pi/4$.

Step 13: Area of first quadrant reachable

$$\text{Area}_{Q1} = \int_0^\pi u \sin u du + \frac{\pi^3}{6} + 2\left(-\frac{\pi}{4}\right) = \pi + \frac{\pi^3}{6} - \frac{\pi}{2} = \frac{\pi}{2} + \frac{\pi^3}{6}$$

Step 14: Total area reachable By symmetry, the region in the second quadrant bounded by the unwrapping string is identical to the first quadrant.

$$\text{Area}_{Q1+Q2} = 2\left(\frac{\pi}{2} + \frac{\pi^3}{6}\right) = \pi + \frac{\pi^3}{3}$$

This gives the total area *including the footprint of the tower*! Wait, the integration gives the area down to the x-axis, which includes the lower half of the tower. We must subtract the area of the tower's base (or half of it, depending on the integration bounds). Actually, the integral calculated the exact area bounded by the curve, the x-axis, and y-axis. The tower is centered at $(0, 1)$ radius 1, so it lives in Q1 and Q2. The integral area includes the semi-circle of the tower. Total area = Area of 2 symmetrical top unwrapped regions + Area of semicircle in lower half ($y < 0$). Area of lower semicircle = $\frac{1}{2}\pi(\pi)^2 = \frac{\pi^3}{2}$. Total reachable area = $2 \times \text{Area}_{Q1} + \frac{\pi^3}{2} - \text{Area of tower}$. Area of tower = $\pi(1)^2 = \pi$.

$$\text{Total Area} = 2\left(\frac{\pi^3}{6} + \frac{\pi}{2}\right) + \frac{\pi^3}{2} - \pi = \frac{\pi^3}{3} + \pi + \frac{\pi^3}{2} - \pi = \frac{5\pi^3}{6}$$

[M1] Suitable strategy to combine components.

[B1] Correct area for semicircle ($\pi^3/2$) or tower (π).

[A1] Correct final total reachable area.
